SPSS 1 Introduction

Learning Outcomes

- By the end of this module, you should be able to:
  1. Formulate engineering and business problems in statistical model for computer-aided analysis
  2. Apply computer-aided statistical analysis to discover hidden patterns and trends in survey data sets
  3. Validate statistical hypothesis on experiment data using computer-aided analysis
  4. Compose analysis result based on the outputs of computer-aided analysis

In this section

- **Structure of this course**
- Statistic Analysis
- SPSS Interface
- Variable Settings and Level of Measurement
- Data Transformation
- Check Data Error Before Data Analysis

Class Schedule

Section (3 hours each)	Topic	Assignment	Date
1	Introduction of SPSS	1. Data Fraud Detection	16/9
2	Descriptive statistics, graphical presentation	1. Summary Table, Cluster Bar Chart, Cluster Line Chart 2. OLAP Cube, Percentiles, Box Plot 3. Histogram, Normal Distribution	23/9
3	T-Test	1. T-test on students' grades in quizzes and final exam	30/9
4	Correlation	1. Multiple correlation study on helping behaviour	7/10
5	Linear regression	1. Multiple regression study on helping behaviour	14/10
6	Analysis of variance (ANOVA)	1. Relationship between years of education, gender, and employment 2. Relationship between working hours, gender, and highest degree	28/10
7	Factor and reliability analysis	1. Factor analysis on questionnaire 2. Reliability analysis on questionnaire	4/11
8	Tutorial/Report writing		11/11
9	Multiple choice test (1-hour)	1. Test 2. Report	18/11

Structure of this course

- Task
  - Class work for appreciation and practice
  - Not marked, but need your participation
- Assignment
  - Assignment for in-class practice and evaluation
  - Marked for final grade
  - You should do it in-class
  - Every section (section 1-7) carry equal weighting
  - 50% of the total mark of the module

Structure of this course

- Assignment submission
  - Name the file name as
    [Student ID]_[Section No.]
    eg. for assignment report of student 12345678d, in section 2: 12345678d_2.doc
  - Store you assignment in D:\student\[Student_ID]\
  - Always leave a backup in your e-mail box, USB drive, etc
  - Do not put the assignment on the desktop and shutdown the machine. The data will be erased after rebooting.
  - Submit by the end of the class, unless otherwise arranged.
  - Late submission or submission without attendance will not be marked.
  - Copying case will be sent to Academic Secretary directly.

Structure of this course

- Test
  - 10 multiple-choice questions (10 minutes to finish)
  - 1-hour duration
  - Open-book test
  - 30% of the total mark of the module
- Report
  - Report on the methods, findings and evaluation
  - Submit right after completion of module
  - 20% of the total mark of the module

In this section

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- **Statistic Analysis**
- SPSS Interface
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- Data Transformation
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Statistic Analysis

- Statistic Analysis Purpose
  - Simplify information and filter noise
  - Find pattern and relationships
  - Find meaning in data
- Statistic Analysis Application
  - Measure the cause and effect
  - Make predictions
  - Make decisions
  - Data mining

Statistical analysis software

- Do statistics analysis without being a mathematician
- SPSS means:
  - Importing/entering data
  - Processing of data
  - Comparison of data
  - Computation of statistical results
  - Presentation of statistical results
- SPSS does not:
  - Automatically give an answer from a pool of data

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- Structure of this course
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- **SPSS Interface**
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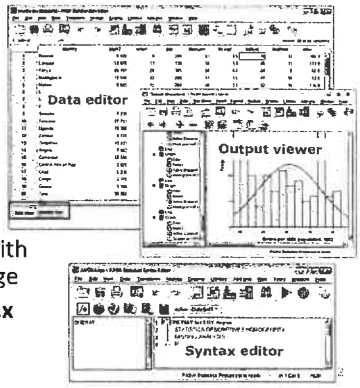
SPSS Interface

1. Data editor
   - Data view
   - Variable view
2. Output viewer
   - Generate report
3. Syntax editor
   - Automate tasks with command language
   - **File > New> Syntax**



SPSS Help

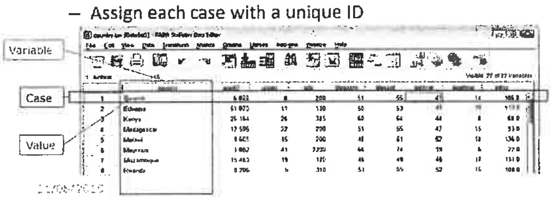
- General help
  - **Help > Topics**
- Dialog box
  - Right click a variable> variable information
- Pivot table
  - Right click row or column header > What's this?

Open a Data File

- Open a data file
  - **File > Open > Data**
    - Select "country.sav" and click **open**
  - **Drag and drop** the data file into the SPSS software
- File formats you can open
  - *.sav (PASW statistical data file)
  - *.xlsx / *.xls (Spreadsheet files)
  - *.txt (delimited/fixed-width data file)

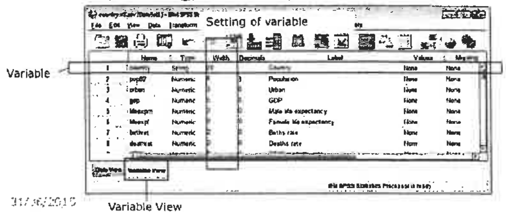
Data View

- Data view is a data table
  - Each row is a Case:
    - The object you want to investigate
  - Each column is a Variable
    - E.g. name, date, age...
    - Use **Variable View** to set the detail of variable
  - Each cell is a Value:
    - Content of a variable of a case
  - Assign each case with a unique ID



Variable View

- Click **Variable View** at the bottom of the **Data View**
- Each row is a variable and its properties
  - E.g. variable name, type, width, decimal, label , value label, missing, measure, ...



Type / String
Type \ Numeric

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Variable Properties

- **Value labels** (Define the value)
  - Add meaning to values
  - e.g. 1: "male", 2: "female"
- **Missing values**
  - Assign value for "no value"
  - e.g. -1 for age
- **Measure** of a variable:
  - 3 Levels of Measurement

Level of measurement

- Measurement can be classified into 3 levels:
  - **Nominal:**
    - categorical data, e.g. 1:blue, 2:red, 3:green
    - no ranking between data
  - **Ordinal:**
    - ordered categorical data, e.g. 1: bad, 2: OK, 3:good
    - usually not scalable: e.g. OK is not twice of bad
  - **Scale** (interval, ratio) :
    - continuous data with comparable values between data,
    - e.g. age of 0-100, population of a city

Level of measurement

- Higher level allows more powerful statistic analysis
  - Scale > Ordinal > Nominal
- Scale data must be Ordinal data; Ordinal data must be Nominal data

Task 1: Import Excel File in SPSS

```
variable 1, variable 2, variable 3, ...
1, 10, 100, ...
2, 20, 200, ...
3, 30, 300, ...
..., ..., ...
```

**.csv file*

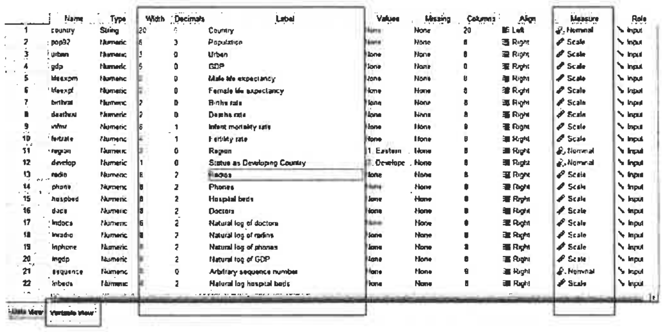
Import

Variable 1	Variable 2	Variable 3	...
1	10	100	...
2	20	200	...
3	30	300	...
...	...	...	...

SPSS file

1. Double click "country.csv" file to check, and close Excel
2. **File > Open > Data**
   - Set **Files of type** to "Text (*txt, *.dat, *.csv)",
   - **Browse** to "country.csv" to open the file and Text Import Wizard
   - Click **Next** > **Select** delimited >
   - **Select** yes (for "Are variable included at the top of your file?")
   - Click **Next > Next > Select** comma and **Deselect** other options
   - Click **Next > Next > Finish**

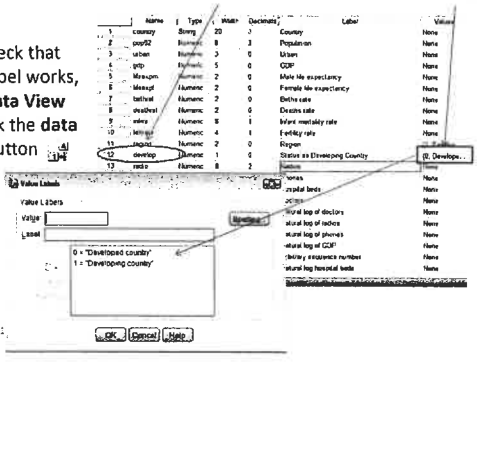
3. Modify the Variables to match the following in Variable View



Step 4 and 5. Value Label

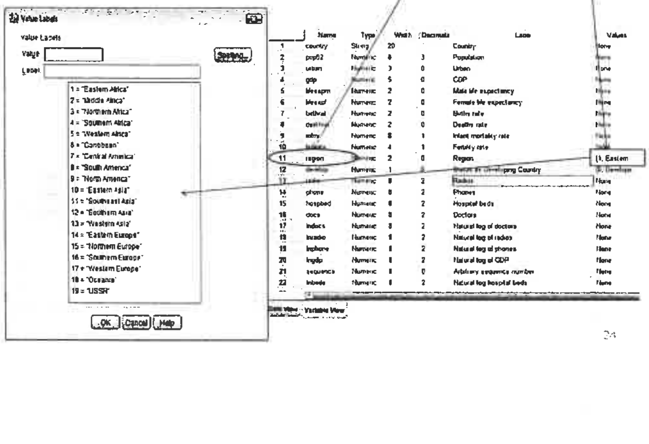
4. Click values column for "12. **develop**" variable and set **Value Labels**:

5. To check that value label works, go to **Data View** and click the **data value** button



Step 6. Value Label

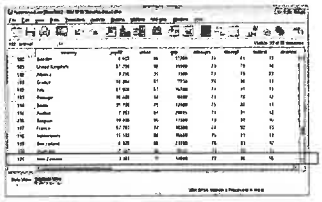
6. In **Variable View**, click **Values** column for "11. **region**" variable and set **Value Labels**:



Step 7 and 8. Add New Case

7. In **Data View**, add one more case at the end of table by typing the following data:

Variable	*value*
country	New Zealand
pop92	3.347
urban	76
gdp	14000
lifeexpm	72
lifeexpf	80
birthrat	15
deathrat	8
infmr	10
fertrate	1.8
region	18
develop	0
radio	90.91
phone	71.43
hospbed	90.09
docs	27.86
lndocs	3.33
lnradio	4.51
lnphone	4.27
lngdp	9.55
sequence	121
lnbeds	4.5



8. **File > Save** to save the file

Task 2: Design of Data File

- The table in the next slide shows the demographical information of a retail customer database
- Design a data file with appropriate name, label, value, measure, etc.
- Key in the data
- Save the file with appropriate filename for future use

Task 2: Design of Data File (con't)

Case	Age	Marital Status	Year in current address	Household Income	Price of primary vehicle	Primary vehicle price category	Level of Education	Years employed	Retired
1	55	Married	12	72	36.2	Luxury	Did not complete high school	23	No
2	56	Unmarried	29	153	76.9	Luxury	Did not complete high school	35	Yes
3	28	Married	9	28	13.7	Economy	Some college	4	No
4	24	Married	4	26	12.5	Economy	College degree	0	No
5	25	Unmarried	2	23	11.3	Economy	High school degree	5	No
6	45	Married	9	76	37.2	Luxury	Some college	13	No
7	42	Unmarried	19	40	19.8	Standard	Some college	10	No
8	35	Unmarried	15	57	28.2	Standard	High school degree	1	No
9	46	Unmarried	26	24	12.2	Economy	Did not complete high school	11	No
10	34	Married	0	89	46.1	Luxury	Some college	12	No
11	55	Married	17	72	35.5	Luxury	Some college	2	Yes
12	28	Unmarried	3	24	11.8	Economy	College degree	4	No
13	31	Married	9	40	21.3	Standard	College degree	0	No
14	42	Unmarried	8	137	68.9	Luxury	Some college	3	No
15	35	Unmarried	8	70	34.1	Luxury	Some college	9	No
16	52	Married	24	159	78.9	Luxury	College degree	16	No
17	21	Married	1	37	18.6	Standard	Some college	0	Yes
18	32	Unmarried	0	28	13.7	Economy	Did not complete high school	2	No
19	42	Unmarried	9	109	54.7	Luxury	Some college	20	No
20	40	Married	12	117	58.3	Luxury	High school degree	19	No

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Data Transformation

- Variables may require pre-processing before running analysis
- For example:
  - take logarithm on the data → e^x → straight line
  - add an offset to the variable
  - reverse the sequence of ordinal data from 0-6 to 6-0

Task: Compute Variable

- To compute new variable from the existing variables:
  1. **Transform** > **Compute Variable**
  2. To take natural logarithm on pop92, **> Function Group > Arithmetic**
  3. **Function and Special Variables** > double click on **Ln >** Double click on <u>pop92</u> from the list of variable, you should see Ln(pop92) in **Numerical Expression**
  4. **Target variable** > input <u>lnpop92</u> (new variable) > **OK**

  Note: you may **If...** to select the cases for computation.

Task: Recode Variable

- Create new variables by dividing existing variable into categories
- eg. classify population into class

pop92	<0	0-10	10-20	20-30	30-40	>40
popclass (new variable)	System-missing	1	2	3	4	5

Task: Recode Variable (Con't)

- To recode <u>popclass</u> from the existing variables <u>pop92</u>:

1. **Transform > Recode Variable into Different Variables**
2. Double click on <u>pop92</u> > **Output Variable** > set **Name as popclass** > set **Label** as **Population Class > Change > Old and New Values**
3. Set **Range, LOWEST through value** to **0 > System-missing > Add** to add the first category (system-missing)
4. Set **Range** as **0** through **10** > set **value** as **1 > Add** to add the second category (1)
5. Repeat 10-20, 20-30, 30-40 for category 2 to 4
6. Set **Range, value through HIGHEST** to **40** > set **value** as **5 > Add** to add the last category (5)
7. **> OK**

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- **Check Data Error Before Data Analysis**

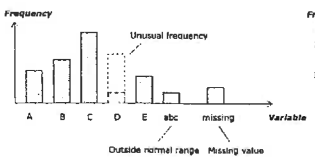
Check Data Value – Data Table

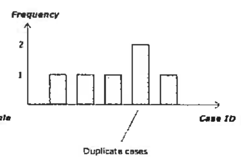
- If data have errors, data analysis would be a waste of time
- Always check data error before data analysis:
  1. Check duplicate cases
  2. Check missing or out-of-scope data
  3. Check distinct values for *nominal/ordinal* data
  4. Check extreme value that are unreasonable
  5. Check distribution of values for unusual frequency
  6. Check relationships of values that are impossible

Country	Population	GDP	Life expectancy of male	Life expectancy of female	Birth rate	Death rate	Infant mortality rate	Fertility rate	Region	Develop
Canada	27.351	-	81	74	14	-7	7.3	1.7	North America	Developed country
United States	256.561	22470	72	79	14	9	10	1.9	Developed country	North America
China	1169.619	360	69	72	22	7	33	2.2	Eastern Asia	Developing country
China	0	19100	77	82	10	100	4	1.8	Eastern Asia	Developed country
1	5	2	6			4			3	

Check Data Value - Frequency

- Always check data error before data analysis
  1. Check duplicate cases
  2. Check missing or out-of-scope data
  3. Check distinct values for *nominal/ordinal* data
  4. Check extreme value that are unreasonable
  5. Check distribution of values for unusual frequency
  6. Check relationships of values that are impossible





Task 2a: Check Data Value

- To check duplicate cases of Norminal/Ordinal data
  1. **Data > Identify Duplicate Cases**
  2. Set **Define Matching Cases By** to <u>Country</u> **> OK**
  3. Look for <u>Primary Case</u> equal to 0 for duplicated case
- To summarize data values
  1. **Analyze > Reports > Case Summaries**
  2. **Select** all required variables **> OK**
- To check distinct values for *nominal/ordinal* data
  1. **Analyze > Descriptive Satistics> Frequency**
  2. Set **Variables** to: <u>region, develop</u> **> OK**

Identify Duplicate Cases

Case ID	Frequency	Percent
Duplicate case	5	5%
Primary case	95	95%
Total	100	100%

Case Summaries

Case ID	Variable 2	Variable 3	Variable 4
Case 1	...	...	...
Case 2	...	...	...
Case 3	...	...	...
Case 4	...	...	...

Frequency table

		Frequency	Percent
Valid	Value 1	80	80%
	Value 2	15	15%
	#?@	3	3%
Missing		2	2%
Total		100	100%

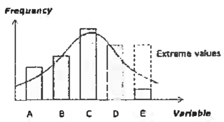
Task 2b: Check Data Distribution for Scale Data

- Check extreme value (instead of all distinct values)
  1. **Analyze > Descriptive Statistic > Explore > Statistic >** check Outliers
  2. Set **Dependent** List as pop92
- Check distribution of values
  1. **Graphs >Legacy Dialog> Histogram**
  2. Set **Variable** to gdp

Extreme values (outliers)

			Case number		Value
Variable	Highest	1		...	1000
		2		...	100
		3		...	99
		4		...	98
		5		...	97
	Lowest	1		...	-3
		2		...	0
		3		...	1
		4		...	2
		5		...	3

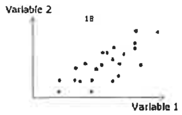
Histogram



Task 2c: Check relationship of values

- To generate scatter plot:
  1. **Graphs > Legacy Dialog > Scatter/Plot > Simple Scatter> Define**
  2. Set **Y Axis** to Infant mortality rate, **X Axis** to Fertility rate > **OK**
  3. **Double click** the plot to edit
  4. Use the **Elements > Data Label Mode** button to **click** a dot to show the case ID
  5. **Right click** the dot to go to the case
  6. What pattern is shown in the scatter plot?
- Try to generate another scatter plot for variables: Phones and Natural log of phones
  – What pattern is shown in the scatter plot?

Scatter plot



Select Cases For Analysis

- To select a subset of cases (eg. gdp>200) for analysis:
  1. **Data > Select Cases >**
  2. Click **If condition is satisfied > If** > set condition to gdp > 200 > **Continue**
  3. Click **Filter out unselected cases** > OK
  – Why is there a new variable "filter_$" after selection?
  – To remove filter: **Data > Select Cases** > Click **All cases**
- To split cases into groups so that each analysis will be repeated for all the groups:
  1. **Data > Split File > Organize Output By Groups**
  2. Then do the analysis

Assignment: Data Fraud Detection

- Import data file country.sav to SPSS
- Data validation
  – Use these statistic tools to check for errors:
    - Identify duplicate case, outliers, case summary, frequency table, histogram, scatter plot
  – Don't need to fix the errors
- Report:
  – The statistic tools used to check the data
  – The statistic results
  – The errors found (in which case and which variable)

Conclusion

- Statistic Analysis is to find pattern and relationships in data
- Always set the correct level of measurement for variables before doing statistics
- Higher level allows more powerful statistic analysis
- Always check data error before data analysis
- Transform data to pre-process prepare data for analysis

Reference

1. "Basic Statistical Analysis" , Richard C. Sprinthall.*Publisher,* Boston, MA : Allyn and Bacon, c2003.
2. "PASW Statistics 18 Statistical Procedures Companion", Marija J. Norušis., Upper Saddle River, NJ : Prentice Hall, c2009
3. "SPSS 17.0 Guide To Data Analysis", Marija J. Norušis, Upper Saddle River, N.J. : Prentice Hall, 2008
4. "IBM SPSS Statistics 19 : Step by Step : a Simple Guide and Reference", Darren George, Paul Mallery, Pearson, 2012
5. "Discovering Statistics using IBM SPSS Statistics", Andy Field, Sage, 2013

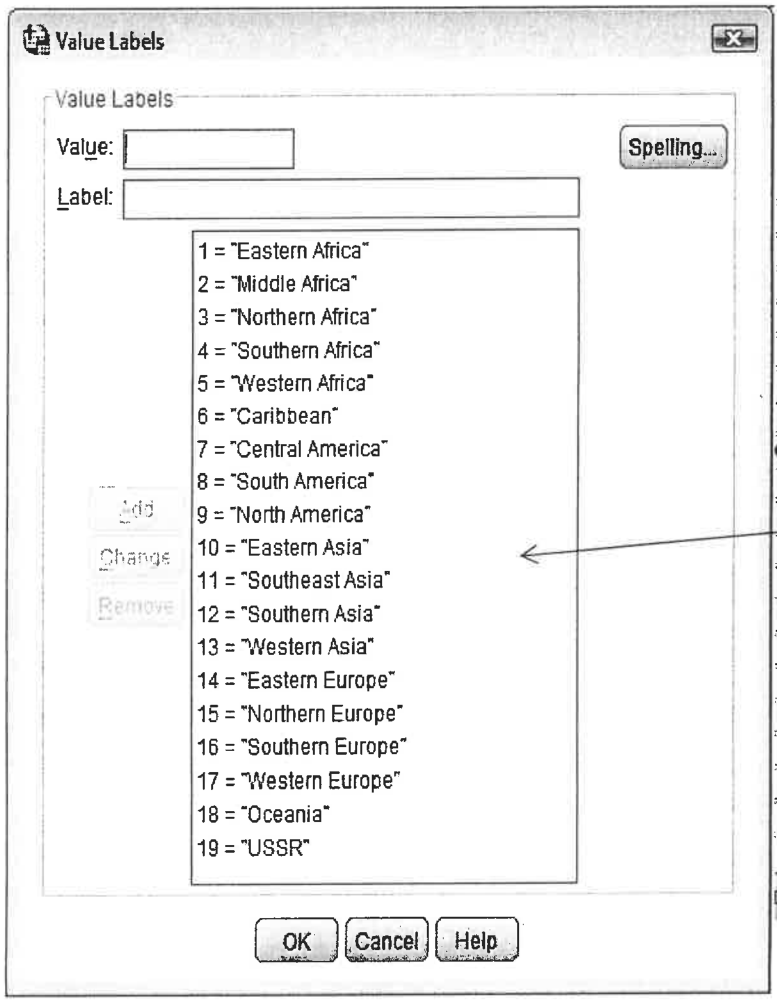
3. Modify the Variables to match the following in Variable View

	Name	Type	Width	Decimals	Label	Values	Missing	Columns	Align	Measure	Role
1	country	String	20	0	Country	None	None	20	Left	Nominal	Input
2	pop92	Numeric	8	3	Population	None	None	8	Right	Scale	Input
3	urban	Numeric	3	0	Urban	None	None	8	Right	Scale	Input
4	gdp	Numeric	5	0	GDP	None	None	8	Right	Scale	Input
5	lifeexpm	Numeric	2	0	Male life expectancy	None	None	8	Right	Scale	Input
6	lifeexpf	Numeric	2	0	Female life expectancy	None	None	8	Right	Scale	Input
7	birthrat	Numeric	2	0	Births rate	None	None	8	Right	Scale	Input
8	deathrat	Numeric	2	0	Deaths rate	None	None	8	Right	Scale	Input
9	infmr	Numeric	8	1	Infant mortality rate	None	None	8	Right	Scale	Input
10	fertrate	Numeric	4	1	Fertility rate	None	None	8	Right	Scale	Input
11	region	Numeric	2	0	Region	{1, Eastern ...	None	8	Right	Nominal	Input
12	develop	Numeric	1	0	Status as Developing Country	{0, Develope...	None	8	Right	Nominal	Input
13	radio	Numeric	8	2	Radios	None	None	8	Right	Scale	Input
14	phone	Numeric	8	2	Phones	None	None	8	Right	Scale	Input
15	hospbed	Numeric	8	2	Hospital beds	None	None	8	Right	Scale	Input
16	docs	Numeric	8	2	Doctors	None	None	8	Right	Scale	Input
17	lndocs	Numeric	8	2	Natural log of doctors	None	None	8	Right	Scale	Input
18	lnradio	Numeric	8	2	Natural log of radios	None	None	8	Right	Scale	Input
19	lnphone	Numeric	8	2	Natural log of phones	None	None	8	Right	Scale	Input
20	lngdp	Numeric	8	2	Natural log of GDP	None	None	8	Right	Scale	Input
21	sequence	Numeric	8	0	Arbitrary sequence number	None	None	8	Right	Nominal	Input
22	lnbeds	Numeric	8	2	Natural log hospital beds	None	None	8	Right	Scale	Input

Data View | Variable View

Step 6. Value Label

6. In **Variable View,** click **Values** column for "11. **region**" variable and set **Value Labels**:



	Name	Type	Width	Decimals	Label	Values
1	country	String	20	0	Country	None
2	pop92	Numeric	8	3	Population	None
3	urban	Numeric	3	0	Urban	None
4	gdp	Numeric	5	0	GDP	None
5	lifeexpm	Numeric	2	0	Male life expectancy	None
6	lifeexpf	Numeric	2	0	Female life expectancy	None
7	birthrat	Numeric	2	0	Births rate	None
8	deathrat	Numeric	2	0	Deaths rate	None
9	infmr	Numeric	8	1	Infant mortality rate	None
10	fertrate	Numeric	4	1	Fertility rate	None
11	region	Numeric	2	0	Region	{1. Eastern ...
12	develop	Numeric	1	0	Status as Developing Country	{0, Develope...
13	radio	Numeric	8	2	Radios	None
14	phone	Numeric	8	2	Phones	None
15	hospbed	Numeric	8	2	Hospital beds	None
16	docs	Numeric	8	2	Doctors	None
17	lndocs	Numeric	8	2	Natural log of doctors	None
18	lnradio	Numeric	8	2	Natural log of radios	None
19	lnphone	Numeric	8	2	Natural log of phones	None
20	lngdp	Numeric	8	2	Natural log of GDP	None
21	sequence	Numeric	8	0	Arbitrary sequence number	None
22	lnbeds	Numeric	8	2	Natural log hospital beds	None

Data View | Variable View

Task 2: Design of Data File (con't)

Case	Age	Marital Status	Year in current address	Household Income	Price of primary vehicle	Primary vehicle price category	Level of Education	Years employed	Retired
1	55	Married	12	72	36.2	Luxury	Did not complete high school	23	No
2	56	Unmarried	29	153	76.9	Luxury	Did not complete high school	35	Yes
3	28	Married	9	28	13.7	Economy	Some college	4	No
4	24	Married	4	26	12.5	Economy	College degree	0	No
5	25	Unmarried	2	23	11.3	Economy	High school degree	5	No
6	45	Married	9	76	37.2	Luxury	Some college	13	No
7	42	Unmarried	19	40	19.8	Standard	Some college	10	No
8	35	Unmarried	15	57	28.2	Standard	High school degree	1	No
9	46	Unmarried	26	24	12.2	Economy	Did not complete high school	11	No
10	34	Married	0	89	46.1	Luxury	Some college	12	No
11	55	Married	17	72	35.5	Luxury	Some college	2	Yes
12	28	Unmarried	3	24	11.8	Economy	College degree	4	No
13	31	Married	9	40	21.3	Standard	College degree	0	No
14	42	Unmarried	8	137	68.9	Luxury	Some college	3	No
15	35	Unmarried	8	70	34.1	Luxury	Some college	9	No
16	52	Married	24	159	78.9	Luxury	College degree	16	No
17	21	Married	1	37	18.6	Standard	Some college	0	Yes
18	32	Unmarried	0	28	13.7	Economy	Did not complete high school	2	No
19	42	Unmarried	9	109	54.7	Luxury	Some college	20	No
20	40	Married	12	117	58.3	Luxury	High school degree	19	No